

The Capacity of Black Holes to Transmit Quantum Information

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- 5 Take-Home Messages

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The Information Paradox

Black holes evaporate to spit *featureless* Hawking radiation. Throw in a pure state and obtain a mixed state. Is information lost? Is QM wrong?

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- Initial understanding of QM allowed only unitary dynamics: pure states evolve to pure states.
- Quantum Shannon Theory: a natural formalism for dealing with information transmission, storage and retrieval.

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What is a Black Hole?

Typical Scenario in Quantum Information

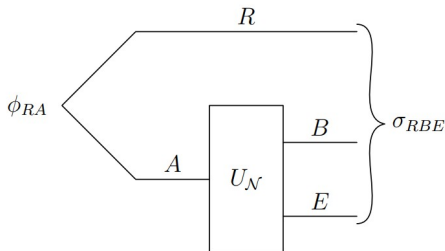
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- A black hole is described by a quantum channel, while its evolution is perfectly *unitary*.

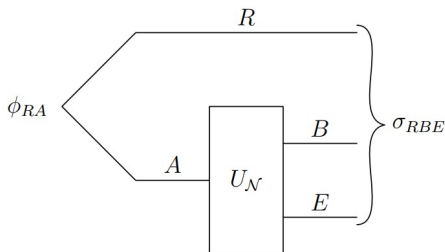


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- Semi-classical model: stationary Schwarzschild black holes as fixed background quantum channels.

Motivation

If the black hole is a channel, what is its *capacity*?

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- If the channel $\mathcal{N}$ is **not too noisy**, then

$$\|\mathcal{D} \circ \mathcal{N}^{\otimes n} \circ \mathcal{E} - \text{id}\|_{\diamond} \leq \epsilon$$

for some $\epsilon, n > 0$, and thus error correction possible as $n \rightarrow \infty$.

Quantum Channels and Capacity

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- The *quantum capacity* $Q(\mathcal{N})$ is then the ratio of the number of qubits we intend to send to the number of qubits the encoder generates to allow for faithful decoding.
- Error-free transmission $\implies Q(\mathcal{N}) > 0$: “**not too noisy**”.

Quantum Channels and Capacity

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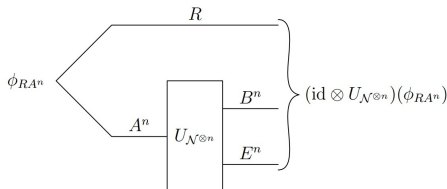
If the black hole is a channel, what is its *capacity*?

- Put formally,

$$Q(\mathcal{N}) = \lim_{n \rightarrow \infty} \frac{1}{n} Q^{(1)}(\mathcal{N}^{\otimes n}).$$

$$Q^{(1)}(\mathcal{N}) \stackrel{\text{def}}{=} \max_{\phi_{RA}} I_c(\mathcal{N}) = \max_{\phi_{RA}} [H(B)_\sigma - H(E)_\sigma].$$

where $H(B)_\sigma$ and $H(E)_\sigma$ are subsystem entropies of B and E on $\sigma \equiv \sigma_{RBE}$.



Hawking Radiation is Featureless

- The Holevo capacity of a black hole is positive, and classical information sent into the black hole escapes as *stimulated emission* and not Hawking radiation.¹

¹Adami, ver Steeg, *Class. Quant. Grav.* 2014

Hawking Radiation is Featureless

- The Holevo capacity of a black hole is positive, and classical information sent into the black hole escapes as *stimulated emission* and not Hawking radiation.¹
- To obtain the quantum channel whose output is Hawking radiation, first consider the channel isometry

$$V(r_\omega) = \prod_{\omega} \exp \left[r_\omega \left(a_k^\dagger b_{-k}^\dagger - a_k b_{-k} \right) \right].$$

where $\tanh r_\omega = \exp(-\pi\omega/\kappa)$, a_k annihilates B's vacuum, b_{-k} annihilates E's vacuum beyond the event horizon and $\kappa = 1/2M$ is the surface gravity.

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Hawking Radiation is Featureless

$$V(r_\omega) = \prod_{\omega} \exp \left[r_\omega \left(a_k^\dagger b_{-k}^\dagger - a_k b_{-k} \right) \right]$$

If vacuum is the initial state,

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To obtain the channel from the isometry,

$$\rho_{\omega, B}^{\text{th}} = \text{Tr}_E (\sigma_{\omega, BE}) = \prod_{\omega} \frac{1}{\cosh^2 r_\omega} \sum_{n=0}^{\infty} \tanh^{2n} r_\omega |n\rangle_B \langle n|$$

By symmetry, $\rho_{\omega, B}^{\text{th}} = \rho_{\omega, E}^{\text{th}}$ and so $Q(\mathcal{N}) = 0 \implies$ Hawking radiation carries no quantum information either!

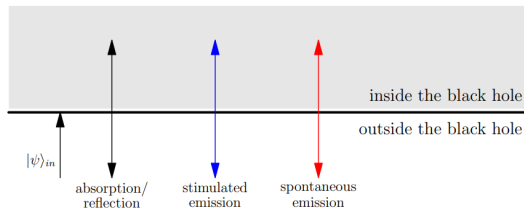
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Black Hole as Transmission Channels

How Does Information Escape?

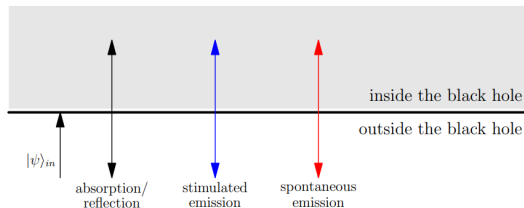
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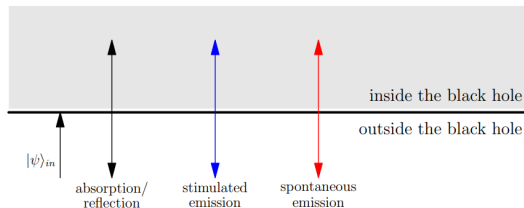


- When Alice throws in $n > 0$ photons, the black hole emits **stimulated radiation**. This is what carries her information out.

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- When Alice throws in $n > 0$ photons, the black hole emits **stimulated radiation**. This is what carries her information out.
- **The Unruh Effect:** A state is prepared by an observer at rest. An observer undergoing *uniform acceleration* will describe the Minkowski vacuum as a *thermal state*. A Black Hole $\Rightarrow$ **Unruh Channel!**

Black Hole $\equiv$ Unruh Channel

Black Hole	Unruh Channel
Alice's infalling photons	Inertial sender
Bob collecting radiation	Uniformly accelerating receiver
Black hole horizon	Rindler horizon
Interior (inaccessible)	Left Rindler wedge ($\xi < 0$)
Stimulated emission	Signal carried to observer
Thermal State: Hawking radiation!	Thermal state seen by observer

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- The channel decomposes as a convex sum of channels (*Direct Sum Channel*)

$$\mathcal{N} = \bigoplus_{\ell=1}^{\infty} p_{\ell} \mathcal{N}_{\ell}, \quad p_{\ell} = \frac{1}{2}(1-z)^2 \ell(\ell+1) z^{\ell-1}$$

where $z = \tanh^2 r_{\Omega} = e^{-2\pi\Omega}$ and $\sum_{\ell} p_{\ell} = 1$.

Cloning Channels and Degradability

- Each subchannel $\mathcal{N}_\ell$ obtained from the decomposition is identified as the optimal $1 \rightarrow \ell$ **quantum cloning channel** $C_{1,\ell}$:

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A channel $\mathcal{N}$ is **degradable** if $\exists$ a channel $\mathcal{D}$ such that $\hat{\mathcal{N}} = \mathcal{D} \circ \mathcal{N}$, where $\hat{\mathcal{N}}$ is the complementary channel. Bob has strictly more information than the environment.

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- Qubit cloning channels $C_{1,\ell}$ are degradable.² Their quantum capacity is therefore single-letter and given by:

$$Q(C_{1,\ell}) = \log_2 \frac{\ell + 1}{\ell}$$

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Quantum Capacity of the Black Hole Channel

- The quantum capacity of a probabilistic mixture of quantum channels is a probabilistic mixture of quantum capacities:

$$\mathcal{N} = \bigoplus_{\ell=1}^{\infty} p_{\ell} C_{1,\ell} \implies Q(\mathcal{N}) = \sum_{\ell=1}^{\infty} p_{\ell} Q(C_{1,\ell})$$

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$$\mathcal{N} = \bigoplus_{\ell=1}^{\infty} p_{\ell} C_{1,\ell} \implies Q(\mathcal{N}) = \sum_{\ell=1}^{\infty} p_{\ell} Q(C_{1,\ell})$$

- Substituting $p_{\ell} = \frac{1}{2}(1-z)^2 \ell(\ell+1)z^{\ell-1}$ and $Q(C_{1,\ell}) = \log_2 \frac{\ell+1}{\ell}$:

$$Q(\mathcal{N}) = \frac{(1-z)^3}{2} \sum_{\ell=1}^{\infty} \ell(\ell+1) z^{\ell-1} \log_2 \frac{\ell+1}{\ell}$$

$$z = \tanh^2 r_{\Omega} = e^{-2\pi\Omega}$$

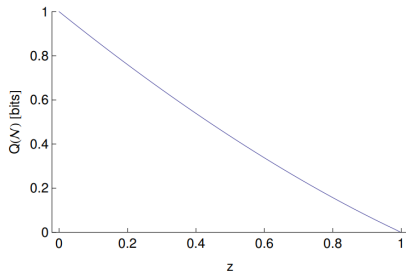


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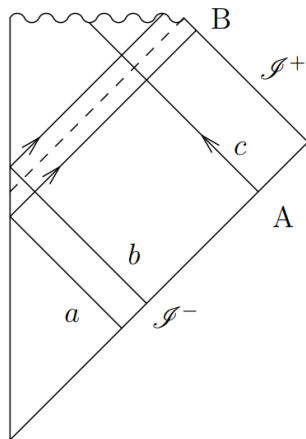
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Tossing Information into the Abyss

Alice throws in her quantum message from outside the black hole .

The annihilation operator for the outgoing radiation mode collected by Bob:

$$a_{\text{out}} = \alpha a - \beta b^\dagger + \gamma c$$



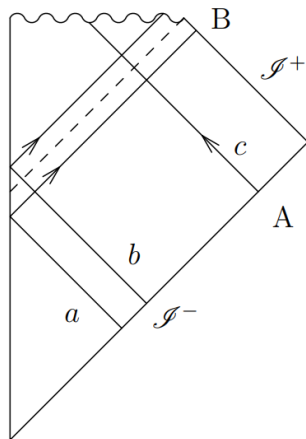
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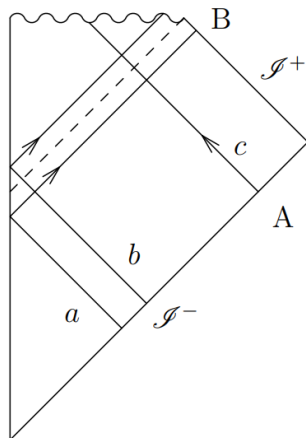
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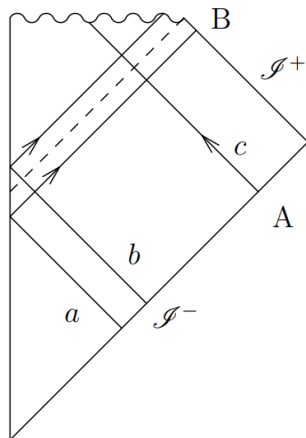
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- **b**: annihilates early-time particles **inside the horizon** $\rightarrow$ the Hawking partner mode, inaccessible to Bob
- **c** annihilates Alice's **late-time input mode** $\rightarrow$ defined after the black hole has formed, this is Alice's quantum message



The Perfectly Reflecting Black Hole

- Let's consider $\alpha = 0$: the **perfectly reflecting** case. The black hole reflects all incoming early-time radiation, none is absorbed.

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- Setting $n = 0$ in the Unruh isometry gives the **two-mode squeezed vacuum**:

$$V_\Omega^{A \rightarrow BE} |0\rangle_A = \frac{1}{\cosh r_\Omega} \sum_{m=0}^{\infty} \tanh^m r_\Omega |m\rangle_B |m\rangle_E$$

mapping it to the earlier Black Hole isometry we discussed.

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- Comparing with the Unruh channel transformations (see appendix), under the identifications $c \leftrightarrow b_\Omega^R$, $b \leftrightarrow b_\Omega^L$:

$$\begin{aligned} \gamma &= \cosh r_\omega & \beta &= \sinh r_\omega & \kappa &\leftrightarrow a \\ \tanh^2 r_\omega &= e^{-2\pi\omega/\kappa} & \longleftrightarrow & \tanh^2 r_\Omega &= e^{-2\pi\omega/a} \end{aligned}$$

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The perfectly reflecting black hole transformation is **exactly** the Unruh transformation under $\kappa \leftrightarrow a$. **The two channels are identical!**

See Derivation

Information is Not Lost!

- Since the perfectly reflecting black hole is exactly the Unruh channel, Alice's information escapes via stimulated emission:

$$\mathcal{N} = \bigoplus_{\ell=1}^{\infty} p_{\ell} C_{h,\ell}, \quad Q(\mathcal{N}) > 0$$

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Entanglement-Breaking Channels

A channel $\mathcal{M}$ is **entanglement-breaking** if $(\text{id}_R \otimes \mathcal{M})(\varphi_{RA})$ is separable for *all* input states φ_{RA} . For such channels:

$$Q(\mathcal{M}) = 0$$

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- The complementary channel, connecting Alice to the black hole interior, is:

$$\hat{\mathcal{N}} = \bigoplus_{\ell=1}^{\infty} p_{\ell} \hat{\mathcal{C}}_{1,\ell}$$

where $\hat{\mathcal{C}}_{1,\ell}$ produces **anti-clones**. This channel is **entanglement-breaking**,³ therefore $Q(\hat{\mathcal{N}}) = 0$.

³K. Brádler, *An infinite sequence of additive channels: the classical capacity of cloning channels*, IEEE Trans. Informat. Theor. **57** (2011) 5497.

Information is Not Lost!

$$\begin{cases} Q(\mathcal{N}) > 0 & \text{Bob receives Alice's imperfect clones, can reconstruct} \\ Q(\hat{\mathcal{N}}) = 0 & \text{interior gets anti-clones, cannot reconstruct} \end{cases}$$

- Alice's quantum information is not destroyed, it escapes via stimulated emission.
- The interior receives only anti-clones, which cannot be used to reconstruct Alice's state.
- **Cloning is forbidden, while information is not lost!**

The Perfectly Absorbing Black Hole

- We implement our Bogoliubov transformation by the Hamiltonian:

$$H_S = ig_\omega(a^\dagger b^\dagger - ab + a^\dagger c - ac^\dagger)$$

$$a_{out} = e^{-iH_S} a e^{iH_S} = a - g_\omega(b^\dagger + c)$$

$$\implies \alpha = 1, \gamma^2 = \beta^2 \equiv g_\omega^2$$

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- After some steps, the final channel from Alice to Bob is:

$$\mathcal{M} = \bigoplus_{\ell=1}^{\infty} p_\ell \mathcal{D}_\ell$$

Information is Lost?

$$\mathcal{M} = \bigoplus_{\ell=1}^{\infty} p_{\ell} \mathcal{D}_{\ell}$$

- For the perfectly absorbing black hole,

$$\mathcal{D}_1(\rho) = \frac{1}{3}\rho + \frac{1}{3}\text{id}_2 \implies \text{entanglement breaking!}$$

See Proof

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See Proof

- It can be shown that $\mathcal{D}_{\ell} \forall \ell$ are **entanglement-breaking**⁴ and so

$$Q(\mathcal{M}) = 0$$

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Information is Lost?

$$\begin{cases} Q(\mathcal{M}) = 0 & \text{Bob receives Alice's imperfect clones, cannot reconstruct} \\ Q(\hat{\mathcal{M}}) > 0 & \text{interior gets clones+anti-clones, can reconstruct} \end{cases}$$

- Complementary behaviour to the perfectly reflecting case.
- **Information is lost**, but that's okay!
- Most importantly, **no-cloning theorem** never violated.

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Summary

With our semiclassical understanding of Schwarzschild black holes,

$\alpha = 0$, Perfect Reflection

$$\begin{cases} Q(\mathcal{N}) > 0 & \text{Bob receives Alice's imperfect clones, can reconstruct} \\ Q(\hat{\mathcal{N}}) = 0 & \text{interior gets anti-clones, cannot reconstruct} \end{cases}$$

$\alpha = 1$, Perfect Absorption

$$\begin{cases} Q(\mathcal{M}) = 0 & \text{Bob receives Alice's imperfect clones, cannot reconstruct} \\ Q(\hat{\mathcal{M}}) > 0 & \text{interior gets clones+anti-clones, can reconstruct} \end{cases}$$

...and the no-cloning theorem is **never violated**! Information loss is not an issue with Quantum Shannon Theory.

Appendix: Derivation of the Mapping

- The Unruh Bogoliubov transformation relating Minkowski modes d_Ω to Rindler modes:

$$\begin{pmatrix} b_\Omega^R \\ b_\Omega^{L\dagger} \end{pmatrix} = \begin{pmatrix} \cosh r_\Omega & \sinh r_\Omega \\ \sinh r_\Omega & \cosh r_\Omega \end{pmatrix} \begin{pmatrix} d_{-\Omega} \\ d_\Omega^\dagger \end{pmatrix}$$

- Inverting (the matrix has $\det = \cosh^2 r_\Omega - \sinh^2 r_\Omega = 1$):

$$d_\Omega = \cosh r_\Omega \cdot b_\Omega^R - \sinh r_\Omega \cdot b_\Omega^{L\dagger}$$

This is the Minkowski mode expressed in terms of Rindler modes: the Unruh channel input-output relation.

- Comparing term by term with the $\alpha = 0$ black hole transformation $a_{\text{out}} = \gamma c - \beta b^\dagger$ under $c \leftrightarrow b_\Omega^R$, $b \leftrightarrow b_\Omega^L$:

$$\gamma = \cosh r_\omega, \quad \beta = \sinh r_\omega$$

- Commutation constraint $[a_{\text{out}}, a_{\text{out}}^\dagger] = 1$:

$$\gamma^2 - \beta^2 = \cosh^2 r_\omega - \sinh^2 r_\omega = 1$$

- Therefore the squeezing parameter:

$$\tanh^2 r_\omega = \frac{\beta^2}{\gamma^2} = \frac{\beta^2}{1 + \beta^2} = e^{-2\pi\omega/\kappa}$$

which maps to $\tanh^2 r_\Omega = e^{-2\pi\omega/a}$ under $\kappa \leftrightarrow a$

Appendix: $Q(\mathcal{M}) = 0$ for Entanglement-Breaking Channels

- The quantum capacity is bounded by the coherent information:

$$I_c(\rho, \Phi) = S(\Phi(\rho)) - S((\Phi \otimes \text{id})(\Psi_\rho))$$

where Ψ_ρ is a purification of the input and S is the von Neumann entropy.

- Separability:** For an entanglement-breaking channel Φ , the joint state $(\Phi \otimes \text{id})(\Psi_\rho)$ is always **separable** (by definition of entanglement-breaking).
- Entropy inequality:** For any separable state σ_{BR} :

$$S(\sigma_B) \leq S(\sigma_{BR})$$

- Therefore:**

$$I_c = \underbrace{S(\Phi(\rho))}_{S_B} - \underbrace{S((\Phi \otimes \text{id})(\Psi_\rho))}_{S_{BR}} \leq 0$$

- Since $I_c \leq 0$ for every input ρ and every n , and $Q \geq 0$ by definition:

$$Q(\Phi) = \lim_{n \rightarrow \infty} \frac{1}{n} \max_{\rho} I_c(\rho, \Phi^{\otimes n}) = 0$$

Appendix: Depolarizing Channel is EB at $q = 2/3$

- A channel Φ is entanglement-breaking iff its Choi state

$$J(\Phi) = (\text{id} \otimes \Phi)(|\Phi^+\rangle\langle\Phi^+|)$$

is separable. For 2×2 , separable $\Leftrightarrow$ PPT.

- For the qubit depolarizing channel

$$\Phi_q(\rho) = (1 - q)\rho + q\frac{I}{2},$$

the Choi state is an isotropic state:

$$J(\Phi_q) = (1 - q)|\Phi^+\rangle\langle\Phi^+| + \frac{q}{4}I.$$

- Take partial transpose:

$$J(\Phi_q)^{T_B} = \frac{1 - q}{2}F + \frac{q}{4}I,$$

where F is the swap operator with eigenvalues $+1$ (symm) and -1 (antisymm).

- Minimum eigenvalue:

$$\lambda_{\min} = \frac{q}{4} - \frac{1 - q}{2} = \frac{3q - 2}{4}.$$

- PPT $\Leftrightarrow \lambda_{\min} \geq 0 \Leftrightarrow q \geq \frac{2}{3}$.

- Hence at $q = \frac{2}{3}$, $J(\Phi_q)$ is PPT $\Rightarrow$ separable $\Rightarrow \Phi_q$ is entanglement-breaking.